

Airframe PID Comparison — Selected Airframes

Airframes: **Ecks 220**, **Ecks 0800 Mod**, **Ecks 0800 Sport**, **Ecks 1kg Mod**, **Ecks 1kg Sport**, **Rok's Quad**, **Ken's 450**, **Ken's Alpha**, **Ken's LadyBug**. Values in unified float format (radians for angles, volts, fractions).

Rate PID Gains

Rate Kp controls roll responsiveness; higher values = more aggressive. Rate Kd provides damping to reduce overshoot.

Param	Ecks 220	Ecks 0800 Mod	Ecks 0800 Sport	Ecks 1kg Mod	Ecks 1kg Sport	Rok's Quad	Ken's 450	Ken's Alpha	Ken's LadyBug
RollRateKp	0.1	0.15	0.24	0.2	0.3	0.095	0.16	0.095	0.16
RollRateKd	0.0045	0.012	0.008	0.006	0.004	0.004	0.006	0.004	0.006

Angle PID Gains

Angle Kp controls attitude stiffness (return-to-level force). Ki eliminates steady-state error in hover. Int Limit caps the integrator windup.

Param	Ecks 220	Ecks 0800 Mod	Ecks 0800 Sport	Ecks 1kg Mod	Ecks 1kg Sport	Rok's Quad	Ken's 450	Ken's Alpha	Ken's LadyBug
RollAngleKp	7	7	8	7	8	5.75	7	5.75	7
RollAngleKi	0.25	0.25	0.2	0.25	0.2	0.15	0.2	0.15	0.1
RollAngleIntLimit	0.002618	0.003	0.005	0.003	0.005	0.002618	0.00733	0.002618	0.005236

Pitch Axis

Pitch rate and angle gains — symmetric to roll by convention on symmetric multirotors.

Param	Ecks 220	Ecks 0800 Mod	Ecks 0800 Sport	Ecks 1kg Mod	Ecks 1kg Sport	Rok's Quad	Ken's 450	Ken's Alpha	Ken's LadyBug
PitchRateKp	0.1	0.18	0.28	0.22	0.32	0.095	0.16	0.095	0.16
PitchRateKd	0.0045	0.012	0.008	0.008	0.005	0.004	0.006	0.004	0.006
PitchAngleKp	7	7	8	7	8	5.75	7	5.75	7
PitchAngleKi	0.25	0.25	0.2	0.25	0.2	0.15	0.1	0.15	0.1
PitchAngleIntLimit	0.002618	0.003	0.005	0.003	0.005	0.002618	0.00733	0.002618	0.005236

Yaw Axis

Yaw rate and angle gains — typically lower Kd than roll/pitch. Yaw angle is a weak hold, often overridden by heading-hold in navigation.

Param	Ecks 220	Ecks 0800 Mod	Ecks 0800 Sport	Ecks 1kg Mod	Ecks 1kg Sport	Rok's Quad	Ken's 450	Ken's Alpha	Ken's LadyBug
YawRateKp	0.1	0.18	0.26	0.15	0.25	0.125	0.2	0.125	0.2
YawRateKd	0.001125	0.008	0.005	0.002	0.002	0.001125	0.001125	0.001125	0.000125
YawAngleKp	3	6	8	4	5	3	3	3	2.25
YawAngleKi	0.25	0.25	0.25	0.25	0.25	0.25	0.25	0.25	0.25
YawAngleIntLimit	0.008727	0.025	0.03	0.015	0.02	0.008727	0.008727	0.008727	0.008727

Navigation Gains

Pos Kp/Ki control position-hold stiffness. Vel Kp controls max correction velocity toward target.

Param	Ecks 220	Ecks 0800 Mod	Ecks 0800 Sport	Ecks 1kg Mod	Ecks 1kg Sport	Rok's Quad	Ken's 450	Ken's Alpha	Ken's LadyBug
NavPosKp	0.2475	0.22	0.28	0.25	0.3	0.2475	0.2475	0.2475	0.33
NavPosKi	0.012	0.012	0.015	0.012	0.015	0.012	0.012	0.012	0.008
NavVelKp	0.3	0.25	0.35	0.3	0.4	0.3	0.6	0.6	0.6

Altitude Gains

Alt Pos Kp/Ki control altitude-hold response. These work on top of the cruise throttle feedforward.

Param	Ecks 220	Ecks 0800 Mod	Ecks 0800 Sport	Ecks 1kg Mod	Ecks 1kg Sport	Rok's Quad	Ken's 450	Ken's Alpha	Ken's LadyBug
AltPosKp	0.5124	0.45	0.5	0.55	0.65	0.4392	0.1464	0.4575	0.915
AltPosKi	0.0046	0.003	0.004	0.003	0.004	0.00276	0.00092	0.00184	0.0138

Cruise & Limits

Cruise throttle is the hover feedforward baseline. Angle/rate limits cap max attitude and rotation speed. Battery capacity and low-voltage threshold set the FC's voltage monitoring.

Param	Ecks 220	Ecks 0800 Mod	Ecks 0800 Sport	Ecks 1kg Mod	Ecks 1kg Sport	Rok's Quad	Ken's 450	Ken's Alpha	Ken's LadyBug
EstCruiseThr	0.54	0.55	0.55	0.6	0.6	0.5	0.31	0.3	0.54
MaxRollAngle	0.523599	0.523599	0.785398	0.523599	0.785398	1.0472	1.0472	1.0472	1.0472
MaxPitchAngle	0.523599	0.523599	0.785398	0.523599	0.785398	1.0472	1.0472	1.0472	1.0472
NavMaxAngle	0.436332	0.436332	0.523599	0.436332	0.523599	0.261799	0.349066	0.349066	0.261799
BatteryCapacity	22	2200	2200	2200	2200	50	22	22	22
LowVoltThres	9.6	9.6	9.6	9.6	9.6	10	13.6	13.8	3
RCChannels	7	7	7	7	7	7	7	7	7
MaxRollRate	10.472	6.28319	10.472	6.28319	10.472	10.472	10.472	10.472	10.472
MaxPitchRate	10.472	3.14159	6.28319	3.14159	6.28319	10.472	10.472	10.472	10.472

Notes

- All values in FC-native units (radians for angles, volts, fractions).
- Ecks 220 is the original default from originaldefaults.h.
- Ecks 0800/1kg variants are manually tuned .af files in the GCS airframes directory.